

AGENTIC AI ENTERPRISE BOOTCAMP

8-Week Production Mastery Program

"Systems ship. Demos don't."

Format	Live + Recorded Saturdays & Sundays
Session Duration	3 Hours per Session (6 hrs/week)
Total Duration	8 Weeks 48 Hours Live Instruction

WHO THIS IS FOR

- ✓ Senior ML Engineers (3+ years) who've prototyped agents but not shipped them
- ✓ Software Architects transitioning to production AI systems
- ✓ Tech Leads building AI strategies for their teams
- ✓ Engineers tired of demos that don't become products
- ✓ Motivated beginners ready to learn production patterns from Day 1

Our Philosophy: Whether you've built 10 agent demos or zero, you'll learn to build agents that actually ship. We teach production patterns throughout—because bad habits are expensive to unlearn.

PHASE 1: FOUNDATIONS & AGENT ARCHITECTURE

Weeks 1-4 | Build strong foundations with production awareness

WEEK 1: Agentic AI Fundamentals & LangGraph Introduction

Session 1 (Saturday) — Mental Models & LangChain Foundations

- What makes something 'Agentic' vs just an LLM call
- Agent Anatomy: Reasoning → Memory → Tools → Loop
- Production Readiness Checklist: Error handling, logging, loop protection
- LangChain Setup: Environment, API keys, secure configuration
- Messages & Conversation Management: SystemMessage, HumanMessage, AIMessage
- Temperature Control: Consistency vs creativity trade-offs

Session 2 (Sunday) — Tools, Agents & LangGraph

- Creating Tools with @tool decorator
- Building Agents with AgentExecutor
- Understanding Agent Architecture and Decision Loops
- AgentExecutor Limitations: Why we need more control
- LangGraph Introduction: Building stateful graphs
- Conditional Routing & State Management

■ **Key Insight:** "Decision loops, not if-else statements. The LLM figures out what to do."

■ **Build:** Customer support workflow with conditional routing (VIP vs Standard paths)

WEEK 2: Production Prompt Engineering & Multi-Agent Systems

Session 1 (Saturday) — Prompting That Ships + Production Foundations

- Advanced Prompt Engineering for Agents
- Prompt Versioning: Git-tracked prompts as code
- A/B Testing Prompts in Production

- Prompt Injection Defenses & Edge Case Handling
- **Production Deep Dive:** Error Handling, Logging & Cost Tracking Patterns
- Implementing try-catch, structured logging, and token budgets

Session 2 (Sunday) — Multi-Agent Collaboration

- Multi-Agent Architecture Patterns
- Agent-to-Agent Communication Protocols
- Error Handling & Retry Logic Across Agents
- Supervisor vs Peer-to-Peer Patterns
- Graceful Degradation: What happens when one agent fails

■ **Key Insight:** *"Prompts are code. Version them. Test them. Deploy them."*

■ **Build:** Multi-agent system with supervisor routing + production error handling

■ **Case Study:** Customer Support Automation — 40% ticket reduction with multi-agent routing

WEEK 3: RAG & Memory Systems for Production Agents

Session 1 (Saturday) — RAG Foundations & Vector Databases

- Why RAG Matters for Production Agents
- Vector Databases Deep Dive: FAISS, PGVector, Redis
- Embedding Strategies & Model Selection
- Chunking Strategies: Why naive chunking fails
- Hybrid Search: Dense + Sparse retrieval for better results

Session 2 (Sunday) — Memory Patterns & Conversation Persistence

- **Memory Architecture:** Buffer, Summary, Token-based, and Entity Memory
- **Session Persistence:** Redis, PostgreSQL, DynamoDB strategies
- Long-term vs Short-term Memory Design
- Retrieval Metrics & Quality Monitoring
- Debugging Retrieval Failures: Why your RAG returns garbage

■ **Key Insight:** *"If you can't measure retrieval quality, you can't improve it."*

■ **Build:** RAG chatbot with persistent memory + retrieval quality dashboard

■ **Case Study:** Enterprise Knowledge Base — How bad chunking cost \$15K in wasted compute

WEEK 4: Tool Design, API Integrations & Agent Testing

Session 1 (Saturday) — Designing Tools LLMs Won't Misuse

- Tool Schema Design Principles
- API Integration Patterns: REST, GraphQL, gRPC
- Authentication & Secrets Management in Production
- Tool Error Feedback to LLM: Making failures recoverable
- **Emerging Standards:** Model Context Protocol (MCP) for tool integration

Session 2 (Sunday) — Production Reliability & Agent Testing

- Rate Limiting & Exponential Backoff
- Fallback Strategies When APIs Fail
- Tool Cost Tracking & Budget Controls
- **Agent Testing Fundamentals:** Unit tests, integration tests for agents
- **Evaluation Basics:** How to measure if your agent is actually working

■ **Key Insight:** *"Every external call will fail. Plan for it. Test for it."*

■ **Build:** Research Assistant with 5+ integrations + automated test suite

■ **Case Study:** The \$40K Failed POC — API failures without fallbacks

PHASE 2: MASTERY, DEPLOYMENT & SHOWCASE

Weeks 5-8 | Scale, deploy, monitor, and showcase production systems

WEEK 5: Advanced LangGraph, Async Patterns & Orchestration

Session 1 (Saturday) — LangGraph Deep Dive

- Why AgentExecutor Isn't Enough for Production
- Complex Pipeline Design with LangGraph
- Branching Logic & Conditional Routing Patterns
- State Persistence Strategies: Redis, PostgreSQL, Memory
- Checkpointing & Recovery: Never lose work mid-pipeline

Session 2 (Sunday) — Async Execution & Human-in-the-Loop

- **Async Patterns:** Running agents without blocking
- **Background Jobs:** Celery, Redis Queue, async task management
- Human-in-the-Loop Patterns: `interrupt_before`, approval gates
- Parallel Execution & Fan-out/Fan-in
- Error Recovery Strategies in Graphs

■ **Key Insight:** *"Graphs give you control. Async gives you scale. Together: production."*

■ **Build:** Async document pipeline with human approval + background processing

■ **Case Study:** Document Processing at Scale — 10x throughput with async patterns

WEEK 6: Deployment, Streaming & Scaling

Session 1 (Saturday) — Containerization & Cloud Deployment

- Containerization with Docker: Best practices for AI workloads
- Deploying on AWS/GCP/Azure: Service selection guide
- Environment Variables & Secrets in Production
- CI/CD Pipelines for AI Agents
- Infrastructure as Code basics

Session 2 (Sunday) — Streaming, APIs & Scaling

- FastAPI + LangServe: Building production endpoints
- **Streaming Responses:** Server-Sent Events (SSE) for real-time UX
- **WebSocket Integration:** Bi-directional agent communication
- Load Balancing & Horizontal Scaling
- Caching Strategies: LLM response caching, embedding caching
- Cost Optimization at Scale

■ **Key Insight:** *"If it's not deployed with streaming, users will think it's broken."*

■ **Build:** Deployed agent API with streaming + public endpoint

■ **Case Study:** From 8s to 800ms perceived latency — The power of streaming

WEEK 7: Guardrails, Monitoring & Enterprise Readiness

Session 1 (Saturday) — Safety, Compliance & Guardrails

- Guardrails for Safety & Compliance
- Input/Output Validation Strategies
- PII Detection & Handling
- Content Moderation & Safety Filters
- Enterprise Compliance Checklist (HIPAA, SOC2, GDPR)

Session 2 (Sunday) — Observability, Evaluation & Optimization

- Logging with Langfuse/LangSmith: Full trace visibility
- Cost Monitoring & Alerting
- Drift Detection & Quality Monitoring

- **Advanced Evaluation:** Building automated eval pipelines
- **Regression Testing:** Catch quality drops before production
- Prompt Versioning & A/B Testing in Production

■ **Key Insight:** *"You can't improve what you can't measure. You can't ship what you can't trust."*

■ **Build:** Production monitoring dashboard with eval pipelines + quality alerts

■ **Case Study:** Enterprise Compliance in Healthcare — HIPAA-compliant agent architecture

WEEK 8: Capstone, Career & Showcase

Session 1 (Saturday) — Capstone Project Briefing

- Capstone Project Requirements & Scope Definition
- Evaluation Criteria: What makes a production-grade submission
- Validation Checkpoints & Milestones
- Building Your AI Portfolio: What to include
- Documentation & Technical Write-up Best Practices
- Timeline: 4 weeks to complete and submit

Session 2 (Sunday) — Resume, Interview Patterns & Career

- Resume Patterns for AI/ML Engineers: What hiring managers look for
- How to Talk About Agent Projects in Interviews
- Common AI Engineer Interview Questions (with answers)
- System Design for Agentic AI: Interview walkthrough
- Career Pathways: IC vs Management in AI
- Q&A; and Next Steps

■ **Key Insight:** *"Your capstone is your proof. Your portfolio is your resume."*

■ **Deliverable:** Capstone project (4-week timeline) + portfolio write-up + interview-ready talking points

PRODUCTION SKILLS COVERAGE

Production Skill	Week Introduced	Depth
Error Handling & Logging	Week 1 (concept) → Week 2 (hands-on)	■■■■■■■
Cost Tracking & Budgets	Week 2 → Week 4 → Week 7	■■■■■■■
Memory & Persistence	Week 3 (dedicated coverage)	■■■■■■■■■
Agent Testing & Evaluation	Week 4 → Week 7 (advanced)	■■■■■■■
Async & Background Jobs	Week 5 (dedicated coverage)	■■■■■■
Streaming Responses	Week 6 (dedicated coverage)	■■■■■
Deployment & CI/CD	Week 6 (dedicated coverage)	■■■■■■■
Monitoring & Observability	Week 7 (dedicated coverage)	■■■■■■■■■
Guardrails & Compliance	Week 7 (dedicated coverage)	■■■■■■■

TECHNOLOGY STACK

Category	Technologies
LLM Libraries	LangChain, LangGraph, LangServe
LLM Providers	OpenAI GPT-4, Anthropic Claude, Google Gemini
Vector Databases	FAISS, PGVector, Redis Vector
Memory & State	Redis, PostgreSQL, DynamoDB
Async & Jobs	Celery, Redis Queue, asyncio
APIs & Streaming	FastAPI, SSE, WebSockets
Deployment	Docker, AWS/GCP, Render, CI/CD
Monitoring	Langfuse, LangSmith, Custom Logging
Testing	pytest, LangSmith Evals, Custom Eval Pipelines
Tools	Python, Git, VS Code/Cursor, MCP

WHAT YOU GET

Live Experience

- 48 hours of live instruction (6 hrs × 8 weeks)
- Recorded sessions with lifetime access
- Live Q&A; during sessions
- Weekly office hours with instructor

Learning Materials

- Complete GitHub repo with all code
- Weekly assignments with solutions
- 3 mini projects + 1 capstone
- Production readiness checklists
- Interactive HTML teaching guides

Community & Support

- Private WhatsApp Community
- Peer discussions and networking
- 1-month post-bootcamp support
- Alumni network access

Career Assets

- Certificate of completion
- 4 portfolio projects (deployed + write-ups)
- Interview preparation materials
- Resume review session

TIME COMMITMENT

Component	Hours/Week
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Live Sessions (Sat + Sun)	6 hours
Hands-on Assignments	3-4 hours
Office Hours (Optional)	1 hour
Total Weekly Commitment	~10 hours

LEARNING OUTCOMES

By Week 4 (Midpoint):

- ✓ Build agents with LangChain and LangGraph
- ✓ Implement production RAG with memory persistence
- ✓ Design reliable tools with proper error handling and testing
- ✓ Understand multi-agent architecture patterns
- ✓ Have 2 working projects with test suites

By Week 8 (Completion):

- ✓ Design and deploy production-grade agent systems with streaming
- ✓ Implement monitoring, guardrails, eval pipelines, and compliance
- ✓ Build async, scalable agent architectures
- ✓ Lead agent projects at your company with confidence
- ✓ Interview confidently for AI Engineer roles
- ✓ Have 4 portfolio projects with live demos and write-ups

OUR GUARANTEES

Quality Guarantee: If you complete all assignments and don't feel confident building production agents, we'll refund 100% of your tuition. **Access Guarantee:** Lifetime access to all recordings, materials, and future updates. Plus free access to one future cohort if you need a refresher.

WEEKLY BUILDS SUMMARY

Week	What You Build
Week 1	Customer support workflow with conditional routing (LangGraph)
Week 2	Multi-agent system with supervisor + production error handling
Week 3	RAG chatbot with persistent memory + retrieval dashboard
Week 4	Research Assistant with 5+ integrations + automated test suite
Week 5	Async document pipeline with human-in-the-loop + background jobs
Week 6	Deployed agent API with streaming + public endpoint
Week 7	Monitoring dashboard with eval pipelines + quality alerts
Week 8	Capstone briefing + Interview prep (4-week project timeline)

READY TO BUILD AGENTS THAT SHIP?

"Systems ship. Demos don't. Learn to build agents that ship." ■